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(57) Abstract :

ABSTRACT DIGITAL TWIN-SYNCHRONIZED RECONFIGURABLE INTELLIGENT SURFACE COMMUNICATION SYSTEM FOR SMART FACTORY ENVIRONMENTS The present invention discloses a Digital Twin-Synchronized Reconfigurable Intelligent Surface (DT-RIS) communication system (100) for smart manufacturing environments operating in 6G contexts. The system comprises a digital twin module (110) for real-time virtual modeling and predictive channel state estimation, a reconfigurable intelligent surface (120) with programmable sub-wavelength reflecting elements for electromagnetic wave manipulation, an access point controller (130) for coordinated transmit power optimization, a sensor network (140) for environmental parameter monitoring, and a predictive RIS configuration unit (150) implementing alternating optimization algorithms. The digital twin module (110) continuously synchronizes physical factory conditions including temperature, humidity, vibration, and object movement to predict short-term channel fluctuations. Based on these predictions, the RIS controller proactively adjusts reflection coefficients for predictive beam steering and phase-shift optimization. The system achieves 41% throughput improvement, 33% latency reduction, 27% enhanced energy efficiency, and 2.5x reliability increase over static RIS configurations, enabling ultra-reliable low-latency communication for Industry 4.0 applications. [to be published with fig.1]

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